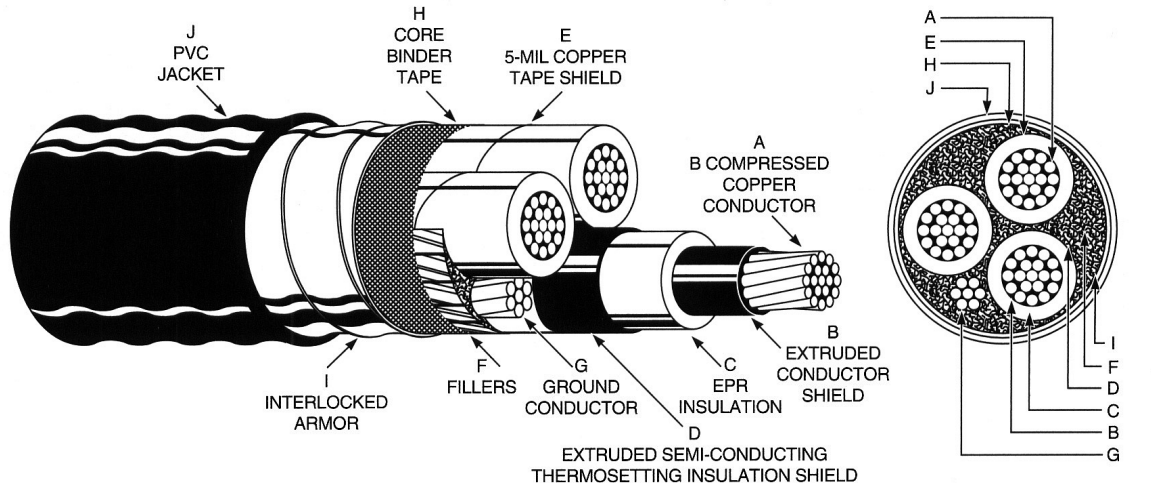
AL13ET

**15kVU Type MV-105 or MC. 3 Copper Conductors.
Thermosetting Conductor and Insulation Shield.
EPR Insulation. Tape Shield.Copper Ground Wire.
Aluminum Armor. PVC Jacket. RoHS**



APPLICATIONS

Southwire AL13ET Type MV-105 or MC Cable is for use in aerial installations, direct burial, concrete-encased installations, cable trays, troughs or continuous rigid cable supports. These cables are capable of operating continuously at a maximum conductor temperature of 105°C for normal operation, 140°C for emergency overload conditions, and 250°C for short circuit conditions, and are rated at 15,000V, 133% insulation level (ungrounded system). Maximum sidewall pressure is 1000 lbs.

SPECIFICATIONS

Southwire AL13ET Type MV-105 or MC Cable is manufactured and tested in accordance with the latest revisions of the following standards and specification:

- UL 1072 - Medium Voltage Power Cables.
- ICEA S-93-639 (NEMA WC 74)
- IEEE 1202/FT4 - Flame Test (70,000 Btu/hr Vertical Tray Test) with CPE and Solonon jacket.
- ICEA T-29-520 - Vertical Cable Tray Flame Test (210,000 Btu/hr).
- NFPA 70 (National Electrial Code)
- AEIC CS-8

CONSTRUCTION

Southwire AL13ET Type MV-105 or MC Cable offers heavy duty construction and flexible EPR insulation. Alternate constructions available. For use in cable trays per NEC Article 392. SOLONON™ low smoke, non halogen polyolefin jackets and CPE jackets are available upon request.

Southwire
One Southwire Drive
Carrollton, Ga. 30119 USA

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RoHS Compliant

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AL13ET 3c, 15kV, 220mil EPR(133%), TS, GW, Alum Interlocked

Product Code	Size	Diameter		0.220" (5.59mm) Insulation Diameter		Extruded Insulation Shield Diameter		Grd. Cond. Size	Armored Diameter		Jacket Thickness		Approx. Overall Diameter		Approx. Net Weight		Allowable Ampacity+	
	AWG or kcmil	inch*	mm	inch**	mm	inch***	mm	AWG	inch	mm	inch	mm	inch	mm	lb./Mft.	kg/km	Direct Burial	In Air
AL13-ET-001	1	.322	8.18	.813	20.64	.868	22.03	4	2.235	56.77	.06	1.52	2.355	59.8	2761	4108	225	210
AL13-ET-010	1/0	.362	9.19	.853	21.65	.908	23.05	4	2.321	58.96	.075	1.91	2.471	62.8	3150	4687	255	240
AL13-ET-030	3/0	.456	11.58	.943	23.94	.998	25.34	3	2.516	63.9	.075	1.91	2.666	67.7	4014	5972	330	315
AL13-ET-250	250	.558	14.17	1.053	26.73	1.108	28.13	2	2.753	69.94	.075	1.91	2.903	73.7	5138	7646	410	400

*Minimum diameter per ASTM standards. **±0.030" ***±0.050
'"*Ampacities are based on the NEC, 2008 Edition. Direct burial ampacities are based on table 310.83 three conductors within an overall covering directly buried, 105°C conductor, 20°C earth ambient temperature. In air ampacities are based on Table 310.71 three conductors within an overall covering in free air, 105°C conductor, 40°C ambient temperature.

SCOPE: This specification covers three conductor EPR (ethylene propylene rubber) insulated, shielded, interlock armored, thermoplastic jacketed power cables with grounding conductor for use in aerial installations, direct burial, concrete-encased installations, cable trays, troughs, or continuous rigid cable supports. These cables are capable of operating continuously at a temperature of 105°C for normal operations, 140°C for emergency overload conditions, and 250°C for short circuit conditions, and are rated at 15,000V, 133% insulation level (ungrounded system).

STANDARDS: The following standards shall form part of this specification: UL Standard 1072 for Medium Voltage Power Cable and ICEA S-93-639 (NEMA WC 74) 5-46 kV Shielded Power Cable for Use in the Transmission and Distribution of Electric Energy.

CONDUCTOR: The conductor shall be Class B compressed concentric stranded bare copper in accordance with ASTM B3 and B8.

CONDUCTOR SHIELD: The conductor shall be shielded with an extruded semi-conducting thermosetting polymeric layer, which shall be firmly bonded to the insulation. The thickness shall be in accordance with the referenced standards.

INSULATION: The insulation shall be EPR (ethylene propylene rubber) meeting the requirements of the referenced standards. The nominal thickness shall be 0.220".

INSULATION SHIELD: The insulation shall be shielded with an extruded semi-conducting thermosetting polymeric layer which shall be identified as semi-conducting. Over this layer shall be applied a helically-wrapped 5-mil copper tape with 25% overlap. The method of phase identification will be similar to ICEA Method 3, using printed circuit numbers and colors (1-BLACK, 2-RED, 3-BLUE).

GROUNDING CONDUCTOR: The ground conductor shall be Class B compressed concentric stranded bare copper in accordance with ASTM B3 and B8.

ASSEMBLY: The insulated and shielded power conductors shall be cabled round with non-wicking, non-hydroscopic fillers and with a grounding conductor in one outer interstice and covered with a binder tape.

ARMOR: A single strip of interlocked armor of aluminum shall be applied over the assembly.

JACKET: The cable shall be covered with a red PVC jacket conforming to the requirements specified for polyvinyl chloride in ICEA. The average thickness shall be in accordance with the referenced standards. The jacket will be sunlight resistant and will meet the requirements of UL 1685 - UL Flame Exposure Test. Optional SOLONON low smoke, non halogen polyolefin jackets and CPE jackets are available upon request.

IDENTIFICATION: Manufacturer's identification shall be printed on the jacket.
TESTS: Qualification tests shall be conducted in accordance with the requirements of AEIC.

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